3 **(a)** the (thermal) energy per unit mass to raise the temperature of a substance by one degree — M1, A1 [2]

(*If ratio not clear for M1 mark, allow 1/2 marks for an otherwise correct answer*)

(b) **(i)** to allow for / determine / cancel heat transfer to / from tube / surroundings — B1 [1]

(*do not allow 'to stop / prevent' heat loss*)

(ii) *either* $P = mc\Delta\theta \pm h$

or $44.9 = 1.58 \times 10^{-3} \times c \times (25.5 - 19.5) \pm h$

or $33.3 = 1.11 \times 10^{-3} \times c \times (25.5 - 19.5) \pm h$ — B1

$(44.9 - 33.3) = (1.58 - 1.11) \times 10^{-3} \times c \times (25.5 - 19.5)$ — C1

$c = 4100\ (4110)\ \mathrm{J\,kg^{-1}\,K^{-1}}$ — A1 [3]

(*allow 1/3 for use of only 33.3 W, 1.11 g s^{-1} leading to 5000 J kg^{-1} K^{-1}*)
(*allow 1/3 for use of only 44.9 W, 1.58 g s^{-1} leading to 4740 J kg^{-1} K^{-1}*)

(c) $V_0 = 27$ *or* $V_{rms} = 19.1$ — C1

$33.3 = 27^2/2R$ *or* $33.3 = 19.1^2/R$ — C1

$R = 11\ \Omega$ — A1 [3]